MICHAEL P. RAGSDALE

Software Architect | Full-Stack Developer & Section 508 Engineer Cloud Infrastructure • Accessibility Engineering • Open-Source Platform Development Norfolk, VA | U.S. Citizen (Clearance Eligible) | michaelpragsdale.com | hireme@michaelpragsdale.com

Professional Summary

Software Architect and Full-Stack Developer specializing in custom web platforms, accessibility-first application architecture, cloud infrastructure, and open-source software development. Creator of the Stardust Engine CMS, a zero-database PHP framework featuring JSON-driven routing, custom SPA navigation, and polyglot decoupled micro-properties across Angular, SvelteKit, React/Next.js, and Vanilla PHP/JS. Experienced across the full software lifecycle, including architecture, development, cloud deployment, DNS management, CI/CD automation, and production operations.

Technical Skills

- **Languages & Frameworks:** PHP 8.5, TypeScript, JavaScript (ES2024), Angular 19, Svelte 5 / SvelteKit, React 19 / Next.js, Python, Bash, SQL, HTML5, Modern CSS3 (`@layer`, `subgrid`, `@container`)
- **Architecture:** MVC Frameworks, Zero-Database JSON/Markdown Data Lakes, SPA Applications, REST APIs
- **Infrastructure:** Ubuntu Linux, Nginx, DigitalOcean (Droplets/Spaces), Cloudflare, Git, CI/CD, FFmpeg
- **Accessibility:** WCAG 2.1 AA/AAA, Federal Section 508, Deque axe, JAWS/NVDA, Dual-Layer Canvas ARIA
- **AI & Automation:** Gemini, LLM Workflows, AI Development Governance, RealESRGAN, Whisper WebVTT

Relevant Technical Experience

Principal Systems Architect & Founder

RaggieSoft Media | Norfolk, VA | raggiesoftmedia.com | Est. 2008 – Present Founded and operate RaggieSoft Media, an independent software development and digital media publishing ecosystem supporting web applications, cloud infrastructure, narrative publishing, music distribution, and open-source platform development.

- **Platform Architecture & CMS:** Architected and open-sourced **Stardust Engine CMS**, an MIT-licensed, zero-database PHP MVC platform utilizing a lightweight JSON data layer, native PHP request routing, and **Elara**, a custom Vanilla JavaScript SPA routing engine for dynamic DOM hydration.
- **Multi-Domain & Cloud Edge:** Orchestrate a production fleet of 9 distinct domains via **Cloudflare DNS/CDN** edge routing, 301 redirection rules, and automated deployment pipelines syncing assets to **DigitalOcean Spaces** (S3-compatible object storage) and Nginx reverse proxies.
- **Automated Media Engineering:** Engineered automated server-side pipelines utilizing **FFmpeg** and Bash to transcode and normalize high-resolution audio assets across multiple distribution formats (FLAC, WAV, MP3, OGG), paired with custom **PHP GD** engines that dynamically composite Open Graph social cards across routes.
- **E-Commerce & Digital Fulfillment:** Integrated **Fourthwall** as an international Merchant of Record (MoR) with custom DNS architecture, powering automated, frictionless global fulfillment of physical merchandise and digital assets without third-party vendor lock-in.
- **AI Governance & Engineering:** Authored comprehensive architectural constraints, repository standards, and prompt protocols to govern AI-assisted engineering workflows across the codebase, ensuring maintainability, code integrity, and architectural consistency.

- **Accessibility by Design:** Enforce strict **WCAG 2.1 AA** and **Section 508** compliance across all UI components, custom modals, media players, and narrative readers, ensuring zero accessibility debt out of the box.

Clerk-Cashier

City of Virginia Beach Parks & Recreation | Virginia Beach, VA | May 2023 – Present

- **Data Engineering & ETL Pipelines:** Architect and manage manual ETL (Extract-Transform-Load) pipelines to ingest, normalize, and secure complex operational and financial data from legacy enterprise platforms (ActiveNet) into highly accessible digital formats.
- **Player-Coach Leadership & Training:** Serve as the primary technical trainer and operational escalation point for municipal staff, translating complex data integrity protocols and Revenue Cycle Management (RCM) compliance into actionable training programs.
- **Stakeholder & Governance Management:** Partner with facility leadership to optimize resource allocation, resolving complex logistical conflicts through data-driven insights and strict adherence to municipal policy.
- **Compliance Enforcement:** Govern output data to guarantee strict Section 508 compliance across all generated operational documents from legacy systems, ensuring institutional adherence to federal accessibility mandates.

Computer Operations

Old Dominion University (ODU Online) | Norfolk, VA | Nov 2019 – Jul 2022

- **Enterprise Accessibility Strategy:** Served as the primary technical gatekeeper for enterprise-level data and web content during a massive, university-wide digital infrastructure migration to the Canvas LMS.
- **Compliance Auditing:** Executed rigorous WCAG 2.1 AA and Section 508 compliance audits, remediating complex codebases (HTML/CSS) and ensuring strict structural integrity across institutional platforms using tools like Deque axe.
- **Architectural Standardization:** Engineered universal, inherently accessible data/architectural templates deployed across the university ecosystem, heavily streamlining data-entry workflows for instructional design faculty.

Mobile Applications Developer Intern

Dominion Enterprises | Norfolk, VA | Sep 2012 – Dec 2012

- Engineered cross-platform mobile applications for iOS and Android utilizing the JavaScript-based Titanium SDK.
- Gained foundational, hands-on experience in mobile architecture and deployment, establishing a decade-long trajectory in software engineering.

Education & Certification

- **Tidewater Community College | Virginia Beach, VA** Associate of Science, Information Technology (Focus: Software Engineering & Web Dev) | Expected Dec 2027
- **Old Dominion University | Norfolk, VA** Bachelor of Science, Customized Studies in Leadership | Prior Coursework
- **Wilson Workforce and Rehabilitation Center (formerly Woodrow Wilson Rehabilitation Center) | Fishersville, VA** Certificate, Business & Information Technology | 2011